



2026/27 SEMESTER 1 – TEST

Course : Academic Support & Enrichment Programme

Unit : ITS105 – Complementary Applied Maths in Computing

Date: *20 July 2026*

Name : _____ Admin No : _____

Unit Group : ITS105-?

Score

100

INSTRUCTION:

- Answer all questions.
- Write your answers in the provided space or type them in a softcopy document.
- If handwritten, scan or convert your answers to PDF/image format.
- Submit all answers to the Brightspace *ITS105 test submission* folder.

Differentiation (30 marks)

1. Given a function $f(x) = (2x^2 + 1)^3$. Find its derivative when $x = 1$.
(10 marks)

2. Given a function $f(x) = (x^3 - x^2 + 1)(x^2 + 2)$. Find its derivative when $x = 1$.
(10 marks)

3. Given a function $f(x) = \frac{x^2-1}{x-2}$. Find its derivative when $x = 0$.
(10 marks)

Integration (30 marks)

4. Find the integral $\int x^6 dx$ (10 marks)

5. Find the integral $\int \frac{1}{\sqrt[3]{x^2}} dx$ (10 marks)

6. Evaluate the definite integral $\int_{-1}^2 (6x + 9x^2 - 8) dx$ (10 marks)

Applications (40 marks)

7. Find the equation of tangent line to the curve $y = x^2 - 4$ at $x = 4$.
(10 marks)

8. Given $y = x^3 - 12x$, find the coordinates of the local minima and local maxima.
(10 marks)

9. Find the area bounded by the curve $y = -x^2 + 4x$ and the x-axis ($y = 0$).

(10 marks)

10. A cyclist moves along a straight line. Its velocity $v(t)$ at time t (in seconds) is given by the function: **$v(t) = 3t^2 - 4t + 5$** where $v(t)$ is in meters per second. Calculate the total distance traveled by the particle during the first 5 seconds of its motion.

Note: the distance travelled D is given by the integral: $D = \int_{t_1}^{t_2} v(t) dt$.

(10 marks)

End of Paper